

100%
#Q1. Write a Function to classify a person's age group: Child, Teen, Adult, senior.

```
def person_age():  
    age1=int(input("Enter age of a person "))  
    if 1<=age1<14:  
        print(f"The person with age {age1} falls under : CHILD")  
    elif 14<=age1<18:  
        print(f"The person with age {age1} falls under : TEEN")  
    elif 18<=age1<=40:  
        print(f"The person with age {age1} falls under : ADULT")  
    elif 40<=age1<=80:  
        print(f"The person with age {age1} falls under : Senior")  
    else :  
        print("invalid age ")  
person_age()
```

#Q3. Write a function to check if a number is within a given range (1-100).

```
def check_range():  
    range1=int(input("Enter the number you want to check"))  
    if 1<=range1<=100 :  
        print(f"The Entered number {range1} falls under the RANGE ")  
    else :  
        print(f"Enter valid number instead of {range1}")  
check_range()
```

#Q4. Write a function to find the sum of numbers from 1 to N.

```
def sum_numbers():  
    num1=int(input("Enter number you want sum "))  
    scor1=(0)  
    for i in range(1,num1+1):  
        scor1=scor1+i  
  
    print(f"The sum of numbers from 1 to {num1} is : {scor1}")  
sum_numbers()
```

#Q5. Write a function to take N users number and find the factorial of all numbers.

```
def fact_numbers():  
    num2=int(input("Enter number you want sum "))  
    scor2=(1)  
    for i in range(1,num2+1):  
        scor2=scor2*i  
        print(f"The Factorial of number {i} is : {scor2}")  
fact_numbers()
```

#Q6. Write a function to print all numbers divisible by 7 but not by 5 from 1-100.

```
def div_num():  
    for i in range(1,101):  
        if i%7==0 and i%5!=0:  
            print(i)  
div_num()
```



#Q7. Print a pattern star like this:

```
*  
**  
*  
**  
def patt_star():  
    for i in range(1,5):  
        if i%2!=0 :  
            print(1*"")  
        else:  
            print(2*"")  
patt_star()
```



#Q9. Write a function to take N user numbers and display even and odd separately and find their sum and average.

```
def calculaton():
```

```
    num5=int(input("Enter number according to you"))
```

```
    odd=[]
```

```
    even=[]
```

```
    sum_odd=(0)
```

```
    sum_even=(0)
```

```
    countodd=(0)
```

```
    counteven=(0)
```

```
    for i in range(1,num5+1):
```

```
        if i%2!=0:
```

```
            countodd+=1
```

```
            sum_odd=i+sum_odd
```

```
            odd.append(i)
```

```
        elif i%2==0:
```

```
            counteven+=1
```

```
            sum_even=i+sum_even
```

```
            even.append(i)
```

```
    averageodd= sum_odd/countodd
```

```
    averageeven= sum_even/counteven
```

```
    print(f"the sum of odd numbers {odd} is :{sum_odd} and average is {averageodd}")
```

```
    print(f"the sum of even numbers {even} is :{sum_even} and average is  
{averageeven}")
```

```
    calculaton()
```



#Q10. Create a function to count the number of vowels and consonants in a string.

String = "The quick brown fox jumps over the lazy dog"

```
def count_vowcon():
```

```
    String = ("The quick brown fox jumps over the lazy dog")
```

```
    countvow=(0)
```

```
    countcon=(0)
```

```
    listvow=[]
```

```
    listcon=[]
```

```
    for i in String:
```

```
        #print(i)
```

```
        if i in "aeiouAEIOU":
```

```
            countvow+=1
```

```
            listvow.append(i)
```

```
        else :
```

```
            countcon+=1
```

```
            listcon.append(i)
```

```
    print(f"The list of vowels is {listvow} and total number of vowels is : {countvow}")
```

```
    print(f"The list of consonats is {listcon} and total number of consonants is :
```

```
{countcon}")
```

```
count_vowncon()
```

#14. Write a function to display cube of numbers from 1-50 in a list.

```
def cube():
```

```
    cubelist1=[]
```

```
    for i in range(1,51):
```

```
        cube1=i**3
```

```
        #print(cube1)
```

```
        cubelist1.append(cube1)
```

```
    print(cubelist1)
```

```
cube()
```

#Q15. Write a function to convert Celsius to Fahrenheit.

```
def cel_ferhnite():
```

```
    num9=int(input("Enter celcius temperature value to convert in fahrenheit"))
```

```
    temp_ferh=((9/5)*num9)+32
```

```
    print(f"the temperature conversion of celsius {num9} to fahrenheit is: {temp_ferh}
```

```
")
```

```
cel_ferhnite()
```